

Homework 5

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MATH 421 – The Theory of Single Variable Calculus

Section 003

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Collaboration Statement

Work compared with Griffin Hailman, with whom brief advice and discussion were exchanged on some problems.

Problem 1 (Ross Chapter 2, 7.4)

Give examples of

- (a) A sequence (x_n) of irrational numbers having a limit $\lim x_n$ that is a rational number.
- (b) A sequence (r_n) of rational numbers having a limit $\lim r_n$ that is an irrational number.

Discussion *Foreword:* I realize now that I did not read and all I had to do is provide examples that did not need proofs. Still, despite 2 hours of sunk cost, I am going to keep this (probably incomplete) proof here because I feel mildly proud of it and hope that, in the case where my subsequent work lacks, I hope this justifies some additional points (pretty please).

This problem asks us to construct two specific kinds of sequences using what we know about rational and irrational numbers.

A *sequence* is simply a function whose domain is the natural numbers (or a tail of them), so each integer n corresponds to a real value s_n . We say that a sequence *converges* to a real number s if for every $\varepsilon > 0$ there exists some N such that $|s_n - s| < \varepsilon$ whenever $n > N$.

Recall that a number is *rational* if it can be expressed as $\frac{m}{n}$ for integers m, n with $n \neq 0$; otherwise it is *irrational*. The set of rational numbers is closed under addition, subtraction, and multiplication. From this it follows that: the product of a nonzero rational number and an irrational number is irrational, and the sum of a rational and an irrational number is irrational.

For *part (a)* we want every term of the sequence to be irrational, but the sequence should approach a rational number as n grows. One natural idea is to start with a rational target,

say 1, and then *perturb* it by a small irrational piece that tends to 0. This ensures each term stays irrational, while the irrational part vanishes in the limit.

For *part (b)* we reverse the situation: every term should be rational, but the sequence should approach an irrational limit. Here we can use the *density of the rationals in the reals*, which guarantees that rationals can approximate any real number—even an irrational one—as closely as we like.

Proof for (a) Let

$$x_n = 1 + \frac{\sqrt{2}}{n}, \quad n \in \mathbb{N}.$$

Since $\sqrt{2}$ is irrational, $\frac{\sqrt{2}}{n}$ must also be irrational: if it were rational, then multiplying by the rational number n would make $\sqrt{2}$ rational, which is impossible. Furthermore, adding a rational number (here 1) to an irrational number produces an irrational number. Hence every x_n is irrational.

Because $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, we have

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 + \frac{\sqrt{2}}{n}\right) = 1 + \sqrt{2} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = 1 + 0 = 1.$$

Therefore the sequence (x_n) consists of irrational numbers and converges to the rational number 1. In other words, (x_n) is irrational for all n , and $\lim x_n = 1$.

Proof for (b) We want to construct a sequence of rational numbers that converges to an irrational number. Since $\sqrt{2}$ is irrational, it is a natural choice for our limit. The idea is to take rational numbers that approximate $\sqrt{2}$ more and more closely.

Let

$$r_n = 1.4, 1.41, 1.414, 1.4142, \dots$$

denote the sequence obtained by truncating the decimal expansion of $\sqrt{2}$ after n digits. Formally, each r_n is a rational number because it has only finitely many decimal places. As we add more digits of $\sqrt{2}$, these rational numbers get arbitrarily close to $\sqrt{2}$.

To express this idea symbolically, we can write

$$|r_n - \sqrt{2}| < \frac{1}{10^n}.$$

Since $\frac{1}{10^n} \rightarrow 0$ as $n \rightarrow \infty$, it follows that

$$\lim_{n \rightarrow \infty} r_n = \sqrt{2}.$$

Therefore, (r_n) is a sequence of rational numbers whose limit is the irrational number $\sqrt{2}$. In summary:

$$r_n \in \mathbb{Q} \text{ for all } n, \quad \text{and} \quad \lim_{n \rightarrow \infty} r_n = \sqrt{2}.$$

Problem 2 (Ross Chapter 2, 8.1c)

Prove the following:

$$\lim_{n \rightarrow \infty} \frac{2n-1}{3n+2} = \frac{2}{3}.$$

Discussion Intuitively, the highest powers of n dominate both the numerator and the denominator: $(2n-1)$ behaves like $2n$ and $(3n+2)$ behaves like $3n$, so the ratio should approach $2n/3n = 2/3$. A tidy way to formalize this with limit laws is to divide numerator and denominator by n , which turns the “lower order” terms into $1/n$ -type pieces that go to 0. If we want an explicit convergence argument, we can also compute the difference from $2/3$ exactly and bound it by ε .

Proof (via limit laws) Write

$$\frac{2n-1}{3n+2} = \frac{2 - \frac{1}{n}}{3 + \frac{2}{n}}.$$

Since $\lim_{n \rightarrow \infty} \frac{1}{n} = \lim_{n \rightarrow \infty} \frac{2}{n} = 0$, we have

$$\lim_{n \rightarrow \infty} \left(2 - \frac{1}{n}\right) = 2 \quad \text{and} \quad \lim_{n \rightarrow \infty} \left(3 + \frac{2}{n}\right) = 3 \neq 0.$$

By the quotient law for limits,

$$\lim_{n \rightarrow \infty} \frac{2n-1}{3n+2} = \frac{\lim(2 - \frac{1}{n})}{\lim(3 + \frac{2}{n})} = \frac{2}{3}.$$

Proof (via Convergence) For every n ,

$$\left| \frac{2n-1}{3n+2} - \frac{2}{3} \right| = \left| \frac{(2n-1)(3) - (3n+2)(2)}{(3n+2)3} \right| = \frac{7}{9n+6}.$$

Given $\varepsilon > 0$, choose

$$N > \frac{1}{9} \left(\frac{7}{\varepsilon} - 6 \right).$$

Then for all $n \geq N$,

$$\left| \frac{2n-1}{3n+2} - \frac{2}{3} \right| = \frac{7}{9n+6} \leq \frac{7}{9N+6} < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} \frac{2n-1}{3n+2} = \frac{2}{3}$.

Problem 3 (Ross Chapter 2, 8.4)

Let (t_n) be a bounded sequence, i.e., there exists M such that $|t_n| \leq M$ for all n , and let (s_n) be a sequence such that $\lim s_n = 0$. Prove $\lim(s_n t_n) = 0$.

Discussion So we are given that $|t_n| \leq M \forall n \in \mathbb{N}$ such that t_n is never going to be greater than M . We are asked to prove that if one sequence (s_n) tends to zero and another sequence (t_n) remains bounded, then their product $(s_n t_n)$ also tends to zero.

The intuition is that (t_n) cannot grow without bound—there is a fixed number M such that $|t_n| \leq M$ for all n . As (s_n) becomes smaller and smaller (approaching 0), multiplying it by any bounded number cannot “blow it up.” In fact, the inequality

$$|s_n t_n| = |s_n| |t_n| \leq M |s_n|$$

captures this relationship precisely. Because $|s_n|$ can be made as small as we wish, $M |s_n|$ can also be made arbitrarily small. Therefore, $(s_n t_n)$ must converge to 0.

Proof Let $\varepsilon > 0$ be given. Since $\lim s_n = 0$, there exists N such that

$$|s_n| < \frac{\varepsilon}{M} \quad \text{for all } n > N.$$

Because (t_n) is bounded, we have $|t_n| \leq M$ for all n . Then, for $n > N$,

$$|s_n t_n| = |s_n| |t_n| \leq M |s_n| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

Thus, by the definition of convergence, $\lim_{n \rightarrow \infty} (s_n t_n) = 0$. □

Problem 4 (Ross Chapter 2, 8.5)

- (a) Consider three sequences (a_n) , (b_n) , and (s_n) such that $a_n \leq s_n \leq b_n$ for all n and $\lim a_n = \lim b_n = s$. Prove $\lim s_n = s$.

This is called the “squeeze lemma.”

- (b) Suppose (s_n) and (t_n) are sequences such that $|s_n| \leq t_n$ for all n and $\lim t_n = 0$. Prove $\lim s_n = 0$.

Solution to (a)

Discussion The idea is: for the same index n , each s_n sits between a_n and b_n . As a_n and b_n both get arbitrarily close to s , the “gap” $[a_n, b_n]$ shrinks to the single point s . There is no room left for s_n to go anywhere else, so s_n must also approach s .

Proof Let $\varepsilon > 0$. Since $\lim a_n = s$, there is N_1 such that $|a_n - s| < \varepsilon$ for all $n > N_1$. Similarly, since $\lim b_n = s$, there is N_2 such that $|b_n - s| < \varepsilon$ for all $n > N_2$. With $N = \max\{N_1, N_2\}$, for all $n > N$ we have

$$s - \varepsilon < a_n \leq s_n \leq b_n < s + \varepsilon,$$

so $|s_n - s| < \varepsilon$. Hence, by the definition of convergence, $\lim s_n = s$. □

Solution to (b)

Discussion The hypothesis $|s_n| \leq t_n$ forces $t_n \geq 0$. If $t_n \rightarrow 0$, then eventually t_n is smaller than any prescribed $\varepsilon > 0$. The squeeze $-t_n \leq s_n \leq t_n$ then pins s_n within ε of 0, so $s_n \rightarrow 0$. (Formally, this is just the squeeze lemma with $a_n = -t_n$, $b_n = t_n$, $s = 0$.)

Proof Let $\varepsilon > 0$. Because $\lim t_n = 0$, there exists N such that $n > N$ implies $t_n < \varepsilon$. For such n we have

$$|s_n - 0| = |s_n| \leq t_n < \varepsilon.$$

Therefore $s_n \rightarrow 0$ by the definition of convergence. $\square$

Problem 5 (Ross Chapter 2, 8.8b)

Prove the following:

$$\lim_{n \rightarrow \infty} [\sqrt{n^2 + n} - n] = \frac{1}{2}.$$

Discussion We want to prove $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \frac{1}{2}$ by the ε - N definition. The obstacle is the square root. A standard trick in these situations is to *rationalize* the expression we must control; i.e., multiply and divide by a suitable conjugate so that the square root disappears from the numerator. Here the target limit is $\frac{1}{2}$, so we study

$$\left| \sqrt{n^2 + n} - n - \frac{1}{2} \right|.$$

If we set $a = \sqrt{n^2 + n}$ and $b = n + \frac{1}{2}$, then $a - b = \frac{a^2 - b^2}{a + b}$, which removes the square root from the numerator. This produces an expression whose size is $\sim \frac{1}{n}$ because the denominator grows linearly with n . Once we bound the difference by a constant multiple of $\frac{1}{n}$, the ε - N argument is routine.

Proof Let $\varepsilon > 0$ be given. Write

$$\sqrt{n^2 + n} - n - \frac{1}{2} = \frac{(\sqrt{n^2 + n} - n - \frac{1}{2})(\sqrt{n^2 + n} + n + \frac{1}{2})}{\sqrt{n^2 + n} + n + \frac{1}{2}} = \frac{(n^2 + n) - (n + \frac{1}{2})^2}{\sqrt{n^2 + n} + n + \frac{1}{2}}.$$

Since $(n + \frac{1}{2})^2 = n^2 + n + \frac{1}{4}$, the numerator simplifies to $-\frac{1}{4}$. Hence

$$\left| \sqrt{n^2 + n} - n - \frac{1}{2} \right| = \frac{1}{4(\sqrt{n^2 + n} + n + \frac{1}{2})}.$$

Because $\sqrt{n^2 + n} \geq n$ for all $n \geq 1$, we have

$$4(\sqrt{n^2 + n} + n + \frac{1}{2}) \geq 4(n + n + \frac{1}{2}) = 8n + 2,$$

and therefore

$$\left| \sqrt{n^2 + n} - n - \frac{1}{2} \right| \leq \frac{1}{8n+2} < \frac{1}{8n} \quad (n \geq 1).$$

Choose $N \in \mathbb{N}$ with $N > \frac{1}{8\varepsilon}$. Then for all $n \geq N$,

$$\left| \sqrt{n^2 + n} - n - \frac{1}{2} \right| < \frac{1}{8n} \leq \frac{1}{8N} < \varepsilon.$$

By the definition of convergence, $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \frac{1}{2}$. □

Problem 6 (Ross Chapter 2, 8.9)

Let (s_n) be a sequence that converges.

- (a) Show that if $s_n \geq a$ for all but finitely many n , then $\lim s_n \geq a$.
- (b) Show that if $s_n \leq b$ for all but finitely many n , then $\lim s_n \leq b$.
- (c) Conclude that if all but finitely many s_n belong to $[a, b]$, then $\lim s_n$ belongs to $[a, b]$.

Solution to (a)

Discussion If a convergent sequence (s_n) is eventually bounded *below* by a , then its tail never dips under a . Intuitively, once s_n is within any $\varepsilon > 0$ of its limit s , all far-out terms lie in $(s - \varepsilon, s + \varepsilon)$. If $s < a$, we can pick ε so small that $(s - \varepsilon, s + \varepsilon)$ lies strictly below a , which contradicts the fact that all sufficiently large s_n are $\geq a$. Hence $s \geq a$.

Proof Let (s_n) converge to s and assume $s_n \geq a$ for all but finitely many n . Suppose, for a contradiction, that $s < a$. Set $\varepsilon := \frac{a - s}{2} > 0$. Because $s_n \rightarrow s$, there exists N_1 such that $|s_n - s| < \varepsilon$ for all $n > N_1$, hence

$$s_n < s + \varepsilon = \frac{s + a}{2} < a \quad (n > N_1).$$

Also, since only finitely many n violate $s_n \geq a$, there exists N_2 with $s_n \geq a$ for all $n > N_2$. Taking $N = \max\{N_1, N_2\}$ yields a contradiction for $n > N$. Therefore $s \geq a$. □

Solution to (b)

Discussion This is the upper-bound analogue of part (a). If $s_n \leq b$ eventually but the limit were $s > b$, then for a small ε the tail $(s - \varepsilon, s + \varepsilon)$ would lie strictly above b , contradicting $s_n \leq b$ for all large n . Hence $s \leq b$.

Proof Let $(s_n) \rightarrow s$ and assume $s_n \leq b$ for all but finitely many n . Suppose, for a contradiction, that $s > b$. Set $\varepsilon := \frac{s-b}{2} > 0$. Then there exists N_1 with $|s_n - s| < \varepsilon$ for all $n > N_1$, so

$$s_n > s - \varepsilon = \frac{s+b}{2} > b \quad (n > N_1).$$

Since only finitely many n violate $s_n \leq b$, there exists N_2 such that $s_n \leq b$ for all $n > N_2$. Taking $N = \max\{N_1, N_2\}$ gives a contradiction for $n > N$. Hence $s \leq b$. $\square$

Solution to (c)

Discussion “All but finitely many s_n lie in $[a, b]$ ” means that beyond some index, $s_n \geq a$ and $s_n \leq b$ simultaneously. Parts (a) and (b) then force $a \leq \lim s_n \leq b$, i.e., the limit belongs to $[a, b]$.

Proof If all but finitely many s_n lie in $[a, b]$, then for some N we have $a \leq s_n \leq b$ for all $n > N$. By (a) we get $\lim s_n \geq a$, and by (b) we get $\lim s_n \leq b$. Therefore $\lim s_n \in [a, b]$. $\square$